

HUMIDIFICATION

Team 3, Thursday, PM section

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INTRODUCTION

Humidification: moisture or water vapor is added to air without changing its dry bulb temperature.

The objective of this experiment is to study the transport of both mass and heat in a wetted wall column and obtain experimental values for the heat and mass transfer coefficients.

Heat and mass transfer coefficients will be obtained from the experimental methods and will be compared to values calculated using the theoretical equation

Independent variables are liquid flowrate, air flowrate, temperature of the incoming air.

THEORY

Humidification is the transfer of water from the liquid phase into a gaseous mix of air and water vapor.

Involves heat and mass transfer between a liquid water and air.

Heat and mass transport are usually described in terms of transport coefficients using the equations for heat and mass flux.

$$q = h \Delta T$$

$$N_A = K_c \Delta c_A$$

Heat flows from the bulk liquid to the interface and from the interface to the bulk gas phase. Latent heat also diffuses into the gas phase from the interface in the water vapor.

The driving forces are the temperature and concentration differences and are linear in these equations

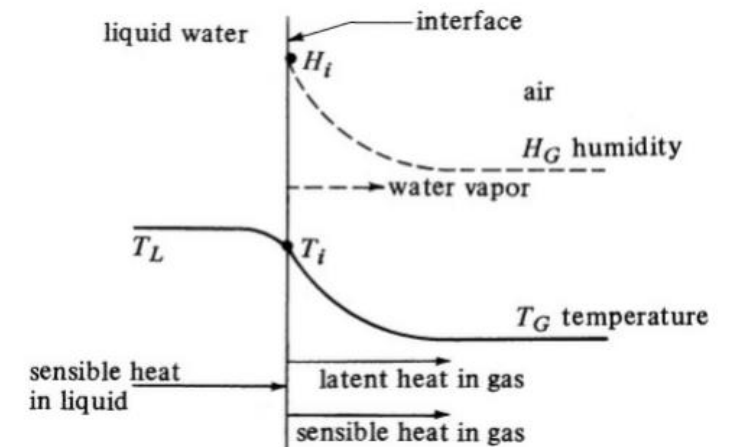


Figure 1. Temperature and concentration profile in upper part of cooling tower

CALCULATION

Calculate k_y , h , and NtG .

- Calculate the experimental mass transfer coefficient.

$$k_{Y,exp} = \frac{(1 - y)_M q_{air}}{Pl}$$

Where

$$k_{Y,exp} = \text{experimental mass transfer coefficient} \left(\frac{\text{lb}}{\text{ft}^2} \cdot \text{s} \right)$$

- Calculate the number of transfer units (NTU).

$$NTU = \ln \left(\frac{Y'_i - Y'_1}{Y'_i - Y'_2} \right) = \frac{k_c Pl}{q_{air}}$$

- Calculate the heat transfer coefficient.

$$h = \frac{q_{air} C_p}{Pl} LMTD$$

Where

$$h = \text{heat transfer coefficient} \left(\frac{\text{Btu}}{\text{ft}^2 \cdot ^\circ\text{F} \cdot \text{s}} \right)$$

$$C_p = \text{heat capacity of air} \left(\frac{\text{Btu}}{\text{lb} \cdot ^\circ\text{F}} \right)$$

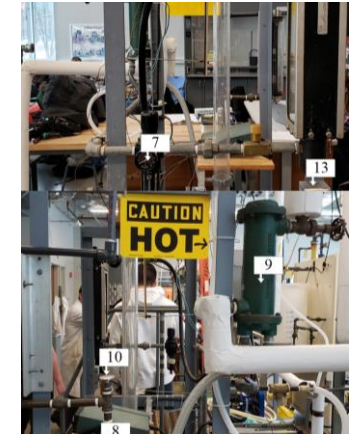
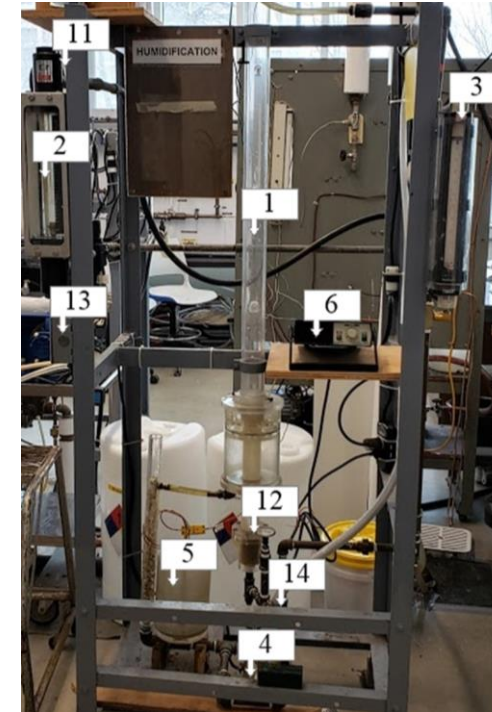
$LMTD$ = Log Mean Temperature Difference ($^\circ\text{F}$)

ΔT_1 = Interface – Inlet temperature ($^\circ\text{F}$)

ΔT_2 = Interface – Outlet temperature ($^\circ\text{F}$)

APPARATUS

- | | |
|------------------------|---------------------|
| 1. Wetted Wall Column | 9. Heat Exchanger |
| 2. Air Rotameter | 10 -12. Thermometer |
| 3. Water Rotameter | 13. Pump Switch |
| 4. Pump | 14. Bypass Valve |
| 5. Reservoir | |
| 6. Thermocouple | |
| 7. Air Pressure Valve | |
| 8. Make Up Water Valve | |



No.	Component
1	Air
2	Water
3	Hygrometer
4	Electricity
5	Low pressure steam
6	Ladder

MATERIAL & SUPPLIES



prelab day



a. Turn on the compressed air tank to allow air flow.



b. Set the compressed air regulator to 40 psig.



c. Set the flowrate of air to 5 SCFM.



d. Use the hygrometer to measure the humidity from the top of the column.



e. Read the wet bulb and dry bulb temperatures on the digital thermometer.

PROCEDURE

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Experiment lab day

1. On Experiment lab day, fill the sump with DI water or tap water, will specify from TA.
2. Turn on the pump to circulate the water to wet the walls of the column.
3. Start airflow at flow rate of 3 SCFM and allow the circulated water to reach a steady state temperature.
4. Measure the steady states temperatures of the air and the water using the wet and dry bulb thermometers. Measure temperature at different locations on the column and record.
5. Measure the steady state air humidity with the hygrometer and record value.
6. Repeated the process two more times.(with 2 other airflow rate

EXPERIMENTAL CHALLENGES



Wait for the stream to reach the steady states



Humidifier Hygrometer reading



Patient, takes forever to heat up the stream



When the stream is on, it's hard to pass the data due to the noise

SAFETY



STEAM HAZARD (HOT!)



PUMP



HIGH TEMPERATURE AIR
STREAM FROM HEAT
EXCHANGER



COMPRESSED AIR



HIGH-PRESSURE SUPPLY



HEAT RESISTANT
GLOVES REQUIRED TO
ADJUST THE HEAT
EXCHANGER

